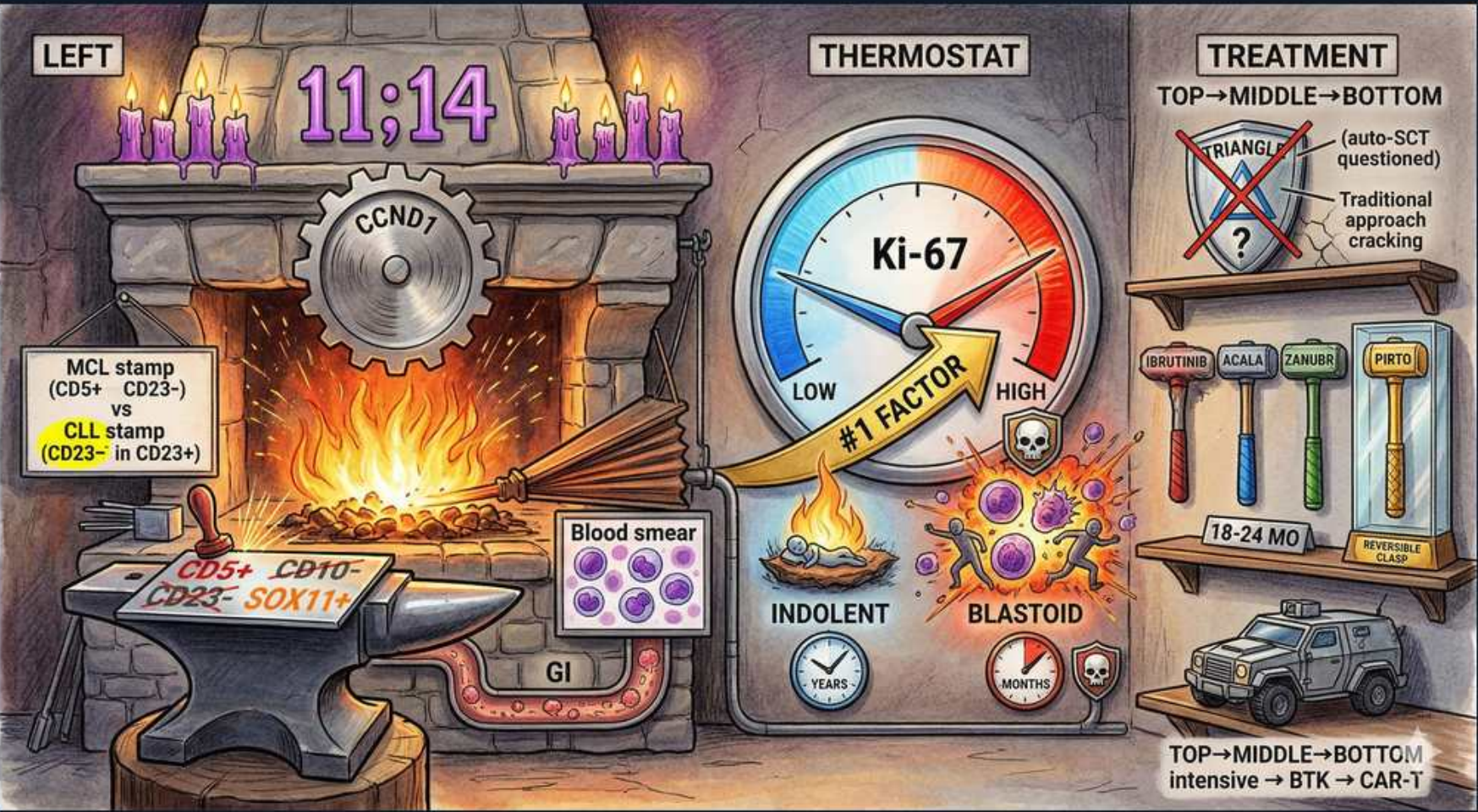


NHL — Mantle Cell Forge (blacksmith forge)



1. t(11;14)

WHAT YOU SEE

Purple '11;14' over the forge candles

WHAT IT ENCODES

Mantle cell lymphoma is defined by t(11;14)(q13;q32), which juxtaposes the CCND1 gene (chromosome 11) with the IgH enhancer (chromosome 14). This is the defining cytogenetic lesion and is present in the great majority of cases. MCL is a mature B-cell NHL that arises from the mantle zone surrounding germinal-center follicles. Mnemonic: Mantle cell = 'eleven-fourteen,' the forge candles count 11 and 14.

BOARD TRAP

WRONG to assign t(11;14)/cyclin D1 to follicular (that is t(14;18)/BCL2) or to Burkitt (that is t(8;14)/MYC). The 11;14 translocation driving cyclin D1 is the mantle-cell signature. A CD5+ B-cell lymphoma with t(11;14) is mantle cell, not CLL.

2. CCND1 gear (cyclin D1)

WHAT YOU SEE

'CCND1' gear above the fire

WHAT IT ENCODES

t(11;14) places CCND1 under the immunoglobulin enhancer, causing overexpression of cyclin D1 (also called BCL1). Cyclin D1 activates CDK4/6 to push cells from G1 into S phase, driving unchecked cell-cycle progression. Nuclear cyclin D1 positivity by immunohistochemistry is the practical diagnostic test for MCL. Mnemonic: cyclin D1 is the gear that spins the cell cycle forward.

BOARD TRAP

WRONG to expect cyclin D1 in follicular (BCL2) or Burkitt (MYC). Cyclin D1 overexpression is essentially specific for mantle cell among the B-cell NHLs (it also appears in hairy cell leukemia and some myeloma). A rare cyclin D1-NEGATIVE MCL exists and is caught by SOX11 positivity.

3. MCL vs CLL stamp

WHAT YOU SEE

'MCL stamp CD5+ CD23-' vs 'CLL stamp CD23+'

WHAT IT ENCODES

Both mantle cell lymphoma and CLL/SLL are CD5-POSITIVE mature B-cell neoplasms, which can confuse them. The key separator is CD23: MCL is CD23-NEGATIVE, CLL/SLL is CD23-POSITIVE. MCL is further defined by cyclin D1/t(11;14) and SOX11; CLL is typically CD23+, CD200+, and FMC7-negative with smudge cells on smear. Mnemonic: both CD5+, but CD23 'sticks' to CLL.

BOARD TRAP

WRONG to call a CD5+ B-cell lymphoma 'CLL' without checking CD23 and cyclin D1 — a CD5+, CD23-NEGATIVE, cyclin D1-positive case is mantle cell, which is far more aggressive and treated entirely differently from indolent CLL. Missing this leads to under-treatment.

4. CD5+ CD10- SOX11+

WHAT YOU SEE

'CD5+ CD10- SOX11+' on blood smear anvil

WHAT IT ENCODES

The classic MCL immunophenotype is CD5-positive, CD10-NEGATIVE, CD23-negative, with cyclin D1 and SOX11 positivity. SOX11 is a transcription factor that confirms MCL even in the rare cyclin D1-negative cases and helps distinguish aggressive (SOX11+) from leukemic non-nodal indolent (SOX11-) MCL. CD10 negativity helps separate MCL from follicular lymphoma (CD10+). Mnemonic: MCL = 'CD5 yes, CD10 no, SOX11 yes.'

BOARD TRAP

WRONG to exclude mantle cell just because cyclin D1 is negative — SOX11 positivity rescues the diagnosis in cyclin D1-negative MCL. Also, CD10 positivity points away from MCL toward follicular lymphoma or Burkitt (both germinal-center, CD10+).

5. GI involvement

WHAT YOU SEE

GI stamped at the base of the anvil where the molten metal pours out across the forge floor — the lymphoma spreading out into the gut as lymphomatous polyposis, the spilled metal flooding the bowel.

WHAT IT ENCODES

Mantle cell lymphoma is frequently disseminated at diagnosis (often stage IV with bone-marrow and peripheral-blood involvement) and characteristically infiltrates the GI tract, producing 'lymphomatous polyposis' — numerous lymphoid polyps throughout the bowel. Many patients have splenomegaly and widespread adenopathy. Mnemonic: MCL forges its way into the gut — think GI polyposis.

BOARD TRAP

Assuming early-stage, localized disease in MCL is the trap — most patients are stage IV at diagnosis with marrow and GI involvement, so staging (including endoscopy when indicated and marrow biopsy) is essential and limited-stage MCL is uncommon.

6. Ki-67 thermostat #1 factor

WHAT YOU SEE

Large 'Ki-67' gauge labeled '#1 FACTOR'

WHAT IT ENCODES

The Ki-67 proliferation index is the single most important PROGNOSTIC biomarker in mantle cell lymphoma: a high Ki-67 (commonly cited cutoff ≥30%) predicts worse survival, while low Ki-67 marks more indolent disease. Ki-67 is combined with the MIPI (Mantle cell International Prognostic Index: age, performance status, LDH, WBC) for risk stratification. TP53 mutation is another powerful adverse marker. Mnemonic: in MCL the thermostat is Ki-67 — the hotter it reads, the worse the outlook.

BOARD TRAP

Using Ki-67 to decide whether a tumor IS mantle cell is the trap — Ki-67 is prognostic, not diagnostic. Diagnosis rests on cyclin D1/t(11;14)/SOX11 and the CD5+/CD23-phenotype; Ki-67 then grades how aggressively that confirmed MCL will behave.

7. Indolent (years)

WHAT YOU SEE

'INDOLENT — YEARS' at low end of gauge

WHAT IT ENCODES

A minority of mantle cell lymphoma is a leukemic, non-nodal, indolent variant — typically SOX11-NEGATIVE, low Ki-67, often presenting with splenomegaly and lymphocytosis but minimal adenopathy. Selected asymptomatic patients can be observed (watch-and-wait) rather than treated immediately, similar to indolent lymphomas. Mnemonic: the cool end of the gauge — SOX11-negative, low Ki-67, watch and wait.

BOARD TRAP

Reflexively starting intensive therapy in every MCL is the trap for the rare indolent leukemic/non-nodal (SOX11-negative, low Ki-67) variant, where observation is appropriate. Conversely, do not observe a SOX11-positive, high-Ki-67, symptomatic MCL — that needs treatment.

8. Blastoid (months)

WHAT YOU SEE

'BLASTOID — MONTHS' skulls at high end

WHAT IT ENCODES

The blastoid and pleomorphic variants of MCL are highly aggressive, with high Ki-67, frequent TP53 mutation, and rapid progression measured in months. They respond poorly to standard chemoimmunotherapy and are prioritized for novel agents and clinical trials. Mnemonic: blastoid = the forge fire at maximum — fast and deadly.

BOARD TRAP

Treating blastoid/TP53-mutated MCL with intensive cytarabine-based chemo and autologous transplant as if it will behave like standard MCL is the trap — these high-risk variants do poorly with chemo and are better served by BTK-inhibitor-based regimens, CAR-T, or trials.

9. Auto-SCT questioned

WHAT YOU SEE

'auto-SCT questioned' with crossed triangle

WHAT IT ENCODES

Historically, fit younger MCL patients received intensive cytarabine-containing induction followed by consolidative AUTOLOGOUS stem-cell transplant. The TRIANGLE trial showed that adding the BTK inhibitor ibrutinib to induction/maintenance improved outcomes and called into question whether autologous transplant is still needed when a BTK inhibitor is incorporated. Mnemonic: the cracked TRIANGLE = TRIANGLE trial questioning auto-SCT.

BOARD TRAP

Assuming autologous transplant is mandatory for all fit MCL patients is now the outdated trap — TRIANGLE data suggest BTK-inhibitor-containing regimens may make upfront auto-SCT unnecessary for many, shifting the frontline paradigm.

10. Traditional approach cracking

WHAT YOU SEE

The old hammers-and-anvil tool shelf on the upper right shown fractured/cracking — the traditional chemo-only frontline anvil splitting apart as BTK inhibitors move to the front.

WHAT IT ENCODES

The chemo-only frontline paradigm for MCL is being replaced by incorporation of BTK inhibitors earlier in the disease course. Frontline and maintenance now increasingly include agents like ibrutinib/acalabrutinib/zanubrutinib (often with rituximab ± chemo), reflecting better efficacy and tolerability than escalating cytotoxic chemotherapy alone. Mnemonic: the old chemo anvil is cracking — BTK inhibitors move to the front.

BOARD TRAP

Defaulting to chemotherapy intensification for relapse instead of a BTK inhibitor is the trap in modern MCL — covalent BTK inhibitors are the cornerstone of relapsed disease and are moving frontline, outperforming repeated cytotoxic chemo in many settings.

11. BTK inhibitors shelf

WHAT YOU SEE

'IBRUTINIB / ACALA / ZANUBR' bottles

WHAT IT ENCODES

Covalent BTK (Bcr/tyrosine kinase) inhibitors block B-cell receptor signaling and are central to relapsed and now frontline MCL: ibrutinib (first-generation, more off-target effects — atrial fibrillation, bleeding, hypertension), and the more selective second-generation acalabrutinib and zanubrutinib (fewer cardiac/bleeding effects). Mnemonic: the BTK shelf — ibrutinib the original, acala/zanu the cleaner refinements.

BOARD TRAP

Continuing ibrutinib despite new atrial fibrillation/bleeding rather than switching to a more selective BTK inhibitor is a management trap — acalabrutinib/zanubrutinib have lower rates of afib and bleeding. Also remember BTK inhibitors raise bleeding risk and interact with anticoagulants.

12. Pirtobrutinib

WHAT YOU SEE

'PIRTO — REVERSIBLE CLAMP'

WHAT IT ENCODES

Pirtobrutinib is a NON-COVALENT (reversible) BTK inhibitor that binds BTK without requiring the C481 cysteine residue. It therefore retains activity in MCL that has become resistant to covalent BTK inhibitors through C481S mutation — a key salvage option after covalent BTK inhibitor failure. Mnemonic: covalent inhibitors grip C481; pirtobrutinib is the reversible clamp that works when C481 is mutated.

BOARD TRAP

Assuming all BTK inhibitors share cross-resistance is the trap — once a covalent BTK inhibitor fails via C481S mutation, switching to ANOTHER covalent agent usually fails too, but the non-covalent pirtobrutinib can still work because it does not depend on C481.

13. 18-24 mo

WHAT YOU SEE

An '18-24 MO' plate fixed between the BTK-inhibitor shelf and the CAR-T truck — marking the short median remission length before MCL relapses and you advance to the next vehicle in the convoy.

WHAT IT ENCODES

Median response durations to successive therapies frame the relapsing-remitting natural history of mantle cell lymphoma. MCL embodies the 'worst of both worlds': it behaves AGGRESSIVELY like a high-grade lymphoma yet is generally INCURABLE with standard chemotherapy like an indolent one, so patients cycle through multiple lines over time. Mnemonic: MCL = aggressive AND incurable — short remissions, repeated relapse.

BOARD TRAP

Telling an MCL patient that aggressive disease means it is curable with chemo is the conceptual trap — unlike Burkitt/DLBCL (aggressive but curable), MCL is aggressive yet typically incurable with conventional chemo, which is why novel agents, transplant, and CAR-T are pursued.

14. Intensive→BTK→CAR-T

WHAT YOU SEE

Armored truck 'intensive → BTK → CAR-T'

WHAT IT ENCODES

A typical MCL treatment sequence: intensive cytarabine-based chemoimmunotherapy induction (for fit patients, historically ± autologous transplant) → BTK inhibitors at relapse (or frontline per TRIANGLE) → brexucabtagene autoleucel (anti-CD19 CAR-T, approved for relapsed/refractory MCL after BTK inhibitor failure). Mnemonic: the armored convoy — induction first, BTK next, CAR-T as the last heavy vehicle.

BOARD TRAP

Skipping straight to CAR-T or chemo without trying a BTK inhibitor in relapse is the sequencing trap — brexucabtagene autoleucel is indicated after BTK-inhibitor failure, and CAR-T carries CRS/ICANS toxicity, so it sits later in the sequence rather than as an early default.